

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1016	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	Sayyad Ahmad Alisha	Reg. No.:	192525415
Date:		Duration:	60 minutes

Code of Conduct

I Sayyad Ahmad Alisha (192525415) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Ahmad

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

Annexure A – Agile Sprint Planning & User Story Workshop

Project Title

Intelligent Agriculture Supply Chain= Traceability Platform

1. Project Vision and Stakeholders

1.1 Project Vision

The **Intelligent Agriculture Supply Chain Traceability Platform** aims to provide a digital system for tracking agricultural products from **farm to consumer**. The system records product origin, harvesting, processing, transportation, storage, and retail information. It helps improve traceability, reduce supply-chain uncertainty, and increase customer confidence.

1.2 Key Stakeholders

Stakeholder	Role
Farmers	Enter crop, farm, and harvesting information.
Collection Centers	Collect products and maintain batch details.
Processors	Record processing and packaging information.
Transporters	Update transportation and delivery details.
Storage Managers	Record storage location and conditions.
Distributors	Manage product distribution.
Retailers	Receive and sell products.
Consumers	View product origin and traceability information.
Administrator	Manage users, products, and system data.
Quality Inspectors	Verify quality and investigate problems.

2. Epics and User Stories

Epic 1: User Management

User Story:

As an administrator, I want to manage user accounts so that only authorized users can access the system.

User Story 1.1

As a farmer, I want to register and log in so that I can add my agricultural product details.

User Story 1.2

As an administrator, I want to approve users so that only valid stakeholders can use the platform.

Epic 2: Product and Harvest Management

User Story:

As a farmer, I want to register my agricultural product and harvesting details so that its origin can be traced.

User Story 2.1

As a farmer, I want to enter crop name, quantity, location, and harvest date so that product information is recorded.

User Story 2.2

As a farmer, I want the system to generate a unique Batch ID so that my product can be tracked throughout the supply chain.

Epic 3: Processing and Transportation

User Story:

As a processor, I want to update processing information so that the product history remains complete.

User Story 3.1

As a transporter, I want to record pickup and delivery details so that product movement can be tracked.

Epic 4: Storage and Traceability

User Story:

As a storage manager, I want to record storage conditions so that product quality can be monitored.

User Story 4.1

As an authorized user, I want to search using a Batch ID so that I can view the complete product journey.

Epic 5: Reports and Consumer Access

User Story:

As an administrator, I want to generate supply-chain reports so that product movement and status can be

analyzed.

User Story 5.1

As a consumer, I want to view product origin and traceability information so that I can trust the product.

3. Acceptance Criteria

User Story: Register Agricultural Product

Given: The farmer is logged into the system.

When: The farmer enters valid product, quantity, location, and harvest details.

Then: The system should save the information and generate a unique Batch ID.

User Story: Track Product

Given: A valid Batch ID exists.

When: The user searches for the Batch ID.

Then: The system should display the available product journey and status.

User Story: Update Processing

Given: A valid batch exists.

When: The processor enters processing details.

Then: The system should update the batch history with the processing information.

User Story: Record Storage

Given: A product batch is stored.

When: The storage manager enters storage details.

Then: The system should save the storage information against the Batch ID.

User Story: Consumer Traceability

Given: A consumer has a valid Batch ID or QR code.

When: The consumer searches or scans it.

Then: The system should display basic product origin and traceability information.

4. Prioritized Product Backlog

Priority	Backlog Item	User Story	Story Points
1 – High	User Login	User can securely log in	3
2 – High	Product Registration	Farmer can register products	5
3 – High	Batch ID Generation	System generates unique Batch ID	3

4 – High	Harvest Details	Farmer records harvesting information	3
5 – High	Batch Tracking	User can track product journey	5
6 – Medium	Processing Details	Processor updates processing information	3
7 – Medium	Transportation	Transporter updates delivery details	5
8 – Medium	Storage Details	Storage manager records conditions	3
9 – Medium	Reports	Admin generates supply-chain reports	5
10 – Low	Consumer View	Consumer views product information	3
11 – Low	QR Code	Consumer scans QR code for tracing	5

5. Sprint Backlog

Sprint 1: User and Product Management

Sprint Goal

To create the basic user management and product registration functionality.

Selected User Stories

1. User registration and login.
2. Admin user approval.
3. Product registration.
4. Harvest information.
5. Batch ID generation.

Sprint Deliverables

At the end of Sprint 1:

- Users can log in.
- Admin can manage users.
- Farmers can register products.
- Harvest details can be recorded.
- A unique Batch ID is generated.

6. Story Point Estimation

Story points are used to estimate the **relative effort and complexity** of each user story.

Fibonacci Scale

1, 2, 3, 5, 8, 13

- **1 point:** Very simple task.
- **2 points:** Small task.
- **3 points:** Moderate task.
- **5 points:** Complex task.
- **8 points:** Very complex task.
- **13 points:** Very large task requiring further breakdown

Example Estimation

User Story	Story Points
User Login	3
Product Registration	5
Harvest Details	3
Batch ID Generation	3
Batch Tracking	5
Processing Details	3
Transportation Tracking	5
Storage Details	3
Report Generation	5
QR Code Tracking	5

7. Sprint Goal and Expected Deliverables

Sprint Goal

To develop the basic foundation of the Intelligent Agriculture Supply Chain Traceability Platform by implementing user management, product registration, harvesting details, and batch identification.

Expected Deliverables

At the end of the sprint, the system should provide:

- Working user login.
- User registration and approval.
- Product registration form.
- Harvest information form.
- Unique Batch ID generation.
- Basic product database.
- Initial testing of the developed modules.

8. Definition of Done

A user story will be considered **Done** when:

- The feature is completely developed.
- Required validation is implemented.
- The feature is tested successfully.
- No major errors are present.
- The feature works with the other modules.
- The completed feature is demonstrated to the stakeholders.

9. Conclusion

The Agile Sprint Planning and User Story Workshop helps organize the development of the **Intelligent Agriculture Supply Chain Traceability Platform** into manageable tasks. The project is divided into epics and user stories, prioritized according to business value, and estimated using story points. Agile Scrum allows the development team to receive continuous stakeholder feedback and improve the system through multiple sprints.